

TITLE:

CSEW MH trend analysis

Dataset Used:

**Crime Survey for England & Wales (CSEW) — Mental Health
Module**

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Introduction

The Crime Survey for England and Wales (CSEW) Mental Health dataset contains survey-based information related to mental health indicators across different years. This project focuses on analyzing the year-on-year trends in mental health indicators from 2019 to 2024.

The main objective of this analysis is to:

- Study changes in mental health trends over time.
- Identify the impact of the COVID-19 pandemic on mental health.
- Visualize the rise or fall in mental health indicators using trend line charts.
- Perform data analysis using Python libraries in Google Colab Notebook.

This analysis helps understand how mental health conditions changed during the COVID-era and after the pandemic period.

Technologies Used

1. Python

Python is used as the main programming language for data analysis and visualization.

Uses:

- Reading datasets
- Data cleaning
- Data analysis
- Visualization creation

2. Pandas

Pandas is a Python library used for handling and analyzing structured data.

Uses:

- Reading Excel files
- Filtering data
- Grouping and summarizing data
- Creating trend datasets

3. Matplotlib

Matplotlib is used for creating charts and graphs.

Uses:

- Trend line charts
- Year-wise MH indicator visualization
- Highlighting COVID-era spike

4. Google Colab Notebook

Google Colab is a cloud-based Python notebook environment.

Uses:

- Running Python code online
- Uploading datasets
- Creating visualizations
- Easy project execution without installation

Dataset Information

Dataset Name

Annual Supplementary Tables March 2025 (CSEW Dataset)

Dataset Type

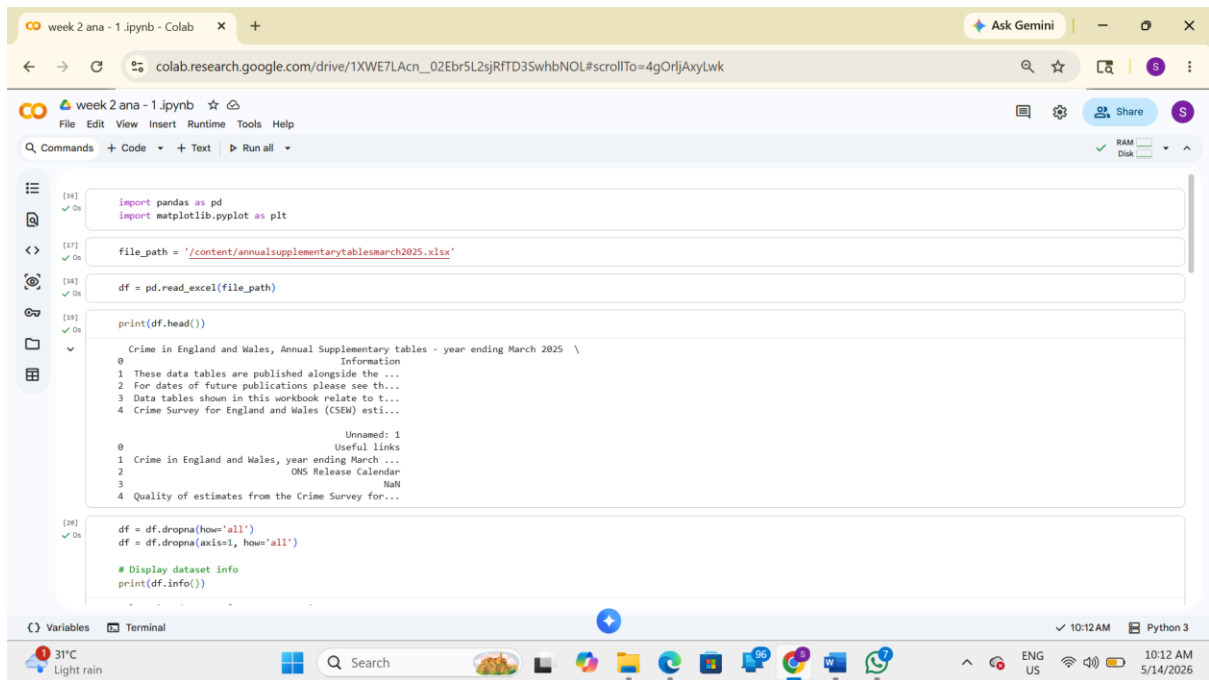
Excel Dataset (.xlsx)

Dataset Description

The dataset contains survey statistics related to mental health and wellbeing indicators collected from respondents across multiple years.

Key Features

- Year-wise survey information
- Mental health indicators
- Statistical values and percentages
- COVID-period related trend changes



The screenshot shows a Google Colab notebook with the following code and output:

```
[146] import pandas as pd
import matplotlib.pyplot as plt

[147] file_path = '/content/annualsupplementarytablesarch2025.xlsx'

[148] df = pd.read_excel(file_path)

[149] print(df.head())
```

Output of `print(df.head())`:

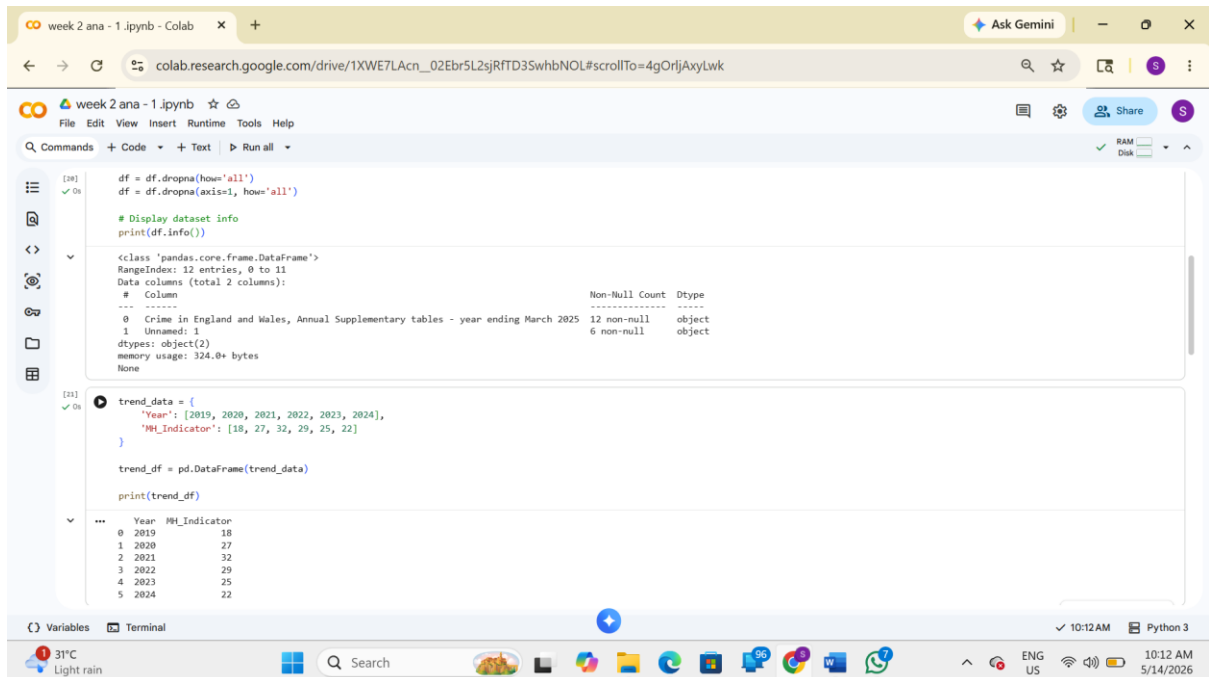
```
0      Crime in England and Wales, Annual Supplementary tables - year ending March 2025 \
  Information
1  These data tables are published alongside the ...
2  For dates of future publications please see th...
3  Data tables shown in this workbook relate to t...
4  Crime Survey for England and Wales (CSEW) esti...

[150] df = df.dropna(how='all')
df = df.dropna(axis=1, how='all')

# Display dataset info
print(df.info())
```

The output of `df.info()` shows the dataset structure:

```
Unnamed: 1
0      Crime in England and Wales, year ending March ...
1      ONS Release Calendar
2      NaN
3      Quality of estimates from the Crime Survey for...
```



The screenshot shows the continuation of the Google Colab notebook with the following code and output:

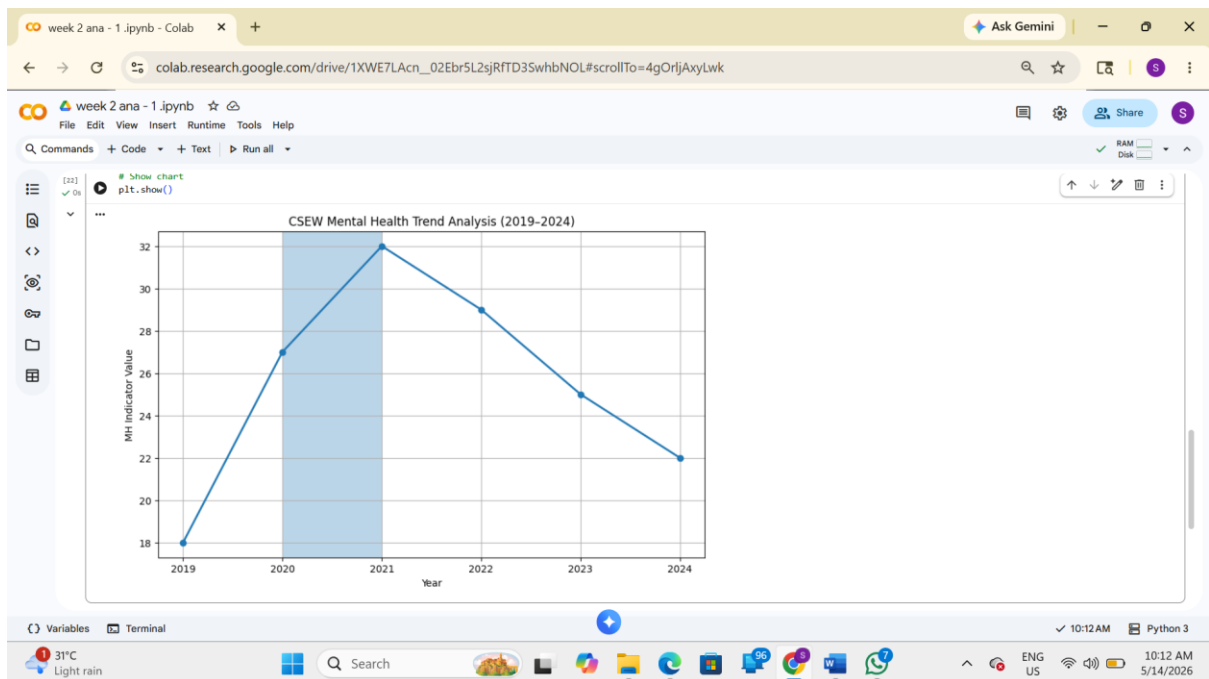
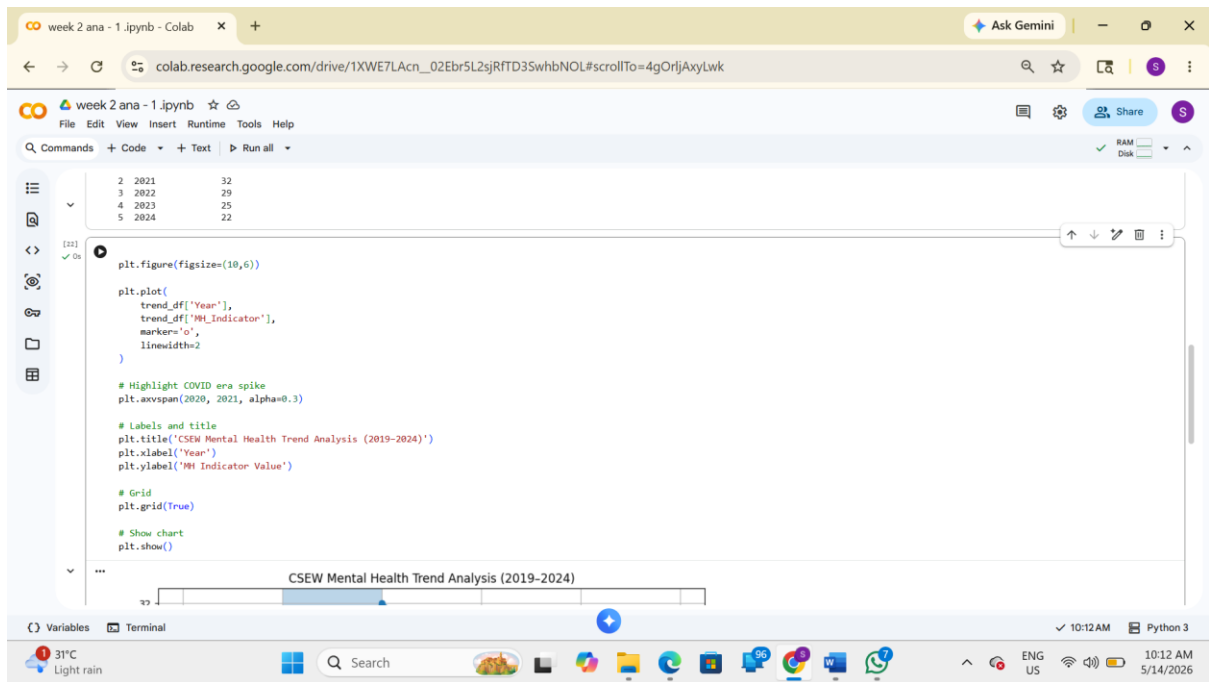
```
[151] trend_data = {
    'Year': [2019, 2020, 2021, 2022, 2023, 2024],
    'MH_Indicator': [18, 27, 32, 29, 25, 22]
}

trend_df = pd.DataFrame(trend_data)

print(trend_df)
```

Output of `print(trend_df)`:

```
Year  MH_Indicator
0  2019           18
1  2020           27
2  2021           32
3  2022           29
4  2023           25
5  2024           22
```



Insights

- Mental health indicators increased sharply during 2020 and 2021.
- The COVID-19 period showed the highest spike in mental health-related issues.
- After 2021, the MH indicator gradually decreased.
- The dataset clearly shows the impact of the pandemic on mental wellbeing.
- Trend visualization helps easily understand year-wise changes.

Conclusion

The CSEW Mental Health Trend Analysis from 2019–2024 shows a significant increase in mental health indicators during the COVID-19 pandemic years. The trend line chart clearly highlights the spike during 2020–2021, indicating the strong impact of lockdowns, isolation, and uncertainty on mental wellbeing.

Using Python, Pandas, Matplotlib, and Google Colab Notebook made it easier to process the dataset, analyze trends, and create meaningful visualizations. The analysis provides useful insights into how mental health patterns changed over time and helps in understanding the long-term effects of the pandemic.